RAPHAEL SULZER

CONTACT

raphael.sulzer.1@gmail.com
raphaelsulzer.de
Nice, France

LinkedIn
GitHub
Google Scholar
ResearchGate

SKILLS

Programming

Python	*9 yrs*
C++	*8 yrs*
Bash	*5 yrs*
HTML/CSS	*4 yrs*
C#	*2 yrs*
JavaScript	*2 yrs*

Languages

German	*Native*
English	*Fluent*
French	*Conversational*
Dutch	*Beginner*

Other

3D modeling · aws · Blender · CGAL · Cesium · CMake · computer vision · conda · deep learning · Docker · GIS · geometry processing · GitHub · LaTeX · LiDAR · machine learning · matplotlib · mesh · nanobind · numpy · photogrammetry · point cloud · pytorch · unix · vscode

EXPERIENCE

RESEARCH ENGINEER
11/2023 - Present
LUXCARTA, MOUANS-SARTOUX, FRANCE
Research and development of 2D and 3D mapping solutions using machine learning and geometry processing techniques applied to remote sensing data.

POSTDOCTORAL RESEARCHER
11/2022 - Present
TITANE, INRIA, SOPHIA-ANTIPOLIS, FRANCE
Research at the intersection of geometry processing, computer vision, and deep learning.

PHD RESEARCHER
12/2018 - 10/2022
LASTIG, INSTITUT GÉOGRAPHIQUE NATIONAL, PARIS, FRANCE
IMAGINE, ÉCOLE DES PONTS PARISTECH, MARNE-LA-VALLÉE, FRANCE
Research at the intersection of geometry processing, computer vision, and deep learning.

LECTURER
12/2018 - 12/2019
ÉCOLE NATIONALE DES SCIENCES GÉOGRAPHIQUES, PARIS, FRANCE
Design and implementation of e-learning courses in photogrammetry and GIS.

GIS DEVELOPER
05/2018 - 07/2019
GEO-COL GIS AND COLLABORATIVE PLANNING, AMSTERDAM, NETHERLANDS
Development of custom GIS applications for public and private customers.

GRADUATE STUDENT INTERN
06/2017 - 02/2018
ARUP, AMSTERDAM, NETHERLANDS
Development of an algorithm for building classification from remote sensing and cadastral data.

EDUCATION

PHD DEGREE, DEEP LEARNING AND GEOMETRY PROCESSING
12/2018 - 10/2022
GUSTAVE EIFFEL UNIVERSITY, MARNE-LA-VALLEÉE, FRANCE
PhD Thesis: Learning Surface Reconstruction from Point Clouds in the Wild (link)

MASTER'S DEGREE, MSC GEOMATICS, CUM LAUDE
08/2016 - 05/2018
DELFT UNIVERSITY OF TECHNOLOGY, DELFT, NETHERLANDS
Master's Thesis: Shape Based Classification of Seismic Building Structural Types (link)

ERASMUS EXCHANGE SEMESTER
08/2015 - 06/2016
DELFT UNIVERSITY OF TECHNOLOGY, DELFT, NETHERLANDS

MSC GEODESY AND GEOINFORMATICS
05/2015 - 01/2016
UNIVERSITY OF STUTTGART, STUTTGART, GERMANY

BACHELOR'S DEGREE, BSC GEODESY AND GEOINFORMATICS
10/2011 - 05/2015
UNIVERSITY OF STUTTGART, STUTTGART, GERMANY
Bachelor's Thesis: Photogrammetric Measurement of Snow Depth Using an UAV Platform (link)

PUBLICATIONS

The P^3 Dataset: Pixels, Points and Polygons for Multimodal Building Vectorization
R. Sulzer, L. Duan, N. Girard, F. Lafarge
2025 preprint arXiv

Concise Plane Arrangements for Low-Poly Surface and Volume Modelling
R. Sulzer, F. Lafarge
2024 European Conference on Computer Vision (ECCV) arXiv

SimpliCity: Reconstructing Buildings with Simple Regularized 3D Models
J.-P. Bauchet, **R. Sulzer**, F. Lafarge, Y. Tarabalka
2024 Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) arXiv

A Survey and Benchmark of Automatic Surface Reconstruction from Point Clouds
R. Sulzer, L. Landrieu, R. Marlet, B. Vallet
2024 IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI) arXiv

Evaluating Surface Mesh Reconstruction Using Real Data
Y. Marchand, L. Caraffa, **R. Sulzer**, E. Clédat, B. Vallet
2023 Photogrammetric Engineering & Remote Sensing (PE&RS) link

Learning Surface Reconstruction from Point Clouds in the Wild
R. Sulzer
2022 Université Gustave Eiffel link

Deep Surface Reconstruction from Point Clouds with Visibility Information
R. Sulzer, L. Landrieu, A. Boulch, R. Marlet, B. Vallet
2022 26th International Conference on Pattern Recognition (ICPR) arXiv

Scalable Surface Reconstruction with Delaunay-Graph Neural Networks
R. Sulzer, L. Landrieu, R. Marlet, B. Vallet
2021 Computer Graphics Forum (CGF) arXiv

Shape Based Classification of Seismic Building Structural Types
R. Sulzer, P. Nourian, M. Palmieri, J. van Gemert
2018 ISPRS International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences link

Track-id: Activity Determination based on Wi-Fi Monitoring
van der Spek, S., Verbree, E., Quak, W., Groeneveld, I. J. D. G., **Sulzer, R.**, Theocharous, E., Xu, Y.
2016 13th International Conference on Location-Based Services (LBS) link

SERVICES

Journal reviewing

- IEEE Transactions on Graphics
- IEEE Transactions on Visualization and Computer Graphics
- ISPRS Journal of Photogrammetry and Remote Sensing
- Computer Graphics Forum

Conference reviewing

- ACM SIGGRAPH
- IEEE Conference on Computer Vision and Pattern Recognition

REFERENCES

Florent LAFARGE

florent.lafarge@inria.fr
TITANE, INRIA
Postdoctoral advisor

Loïc LANDRIEU

loic.landrieu@enpc.fr
IMAGINE, ÉCOLE DES PONTS PARISTECH
Doctoral advisor